some reassurance to know that there are a hundred trillion more of them in the Sun's vast treasure-house of worlds.

THERE ARE ABOUT 200 known asteroids whose paths take them near the Earth. They are called, appropriately enough, "near-Earth" asteroids. Their detailed appearance (like that of their main-belt cousins) immediately implies that they are the products of a violent collisional history. Many of them may be the shards and remnants of once-larger worldlets.

With a few exceptions, the near-Earth asteroids are only a few kilometers across or smaller, and take one to a few years to make a circuit around the Sun. About 20 percent of them, sooner or later, are bound to hit the Earth—with devastating consequences. (But in astronomy, "sooner or later" can encompass billions of years.) Cicero's assurance that "nothing of chance or hazard" is to be found in an absolutely ordered and regular heaven is a profound misperception. Even today, as Comet Shoemaker-Levy 9's encounter with Jupiter reminds us, there is routine interplanetary violence, although not on the scale that marked the early history of the Solar System.

Like main-belt asteroids, many near-Earth asteroids are rocky. A few are mainly metal, and it has been suggested that enormous rewards might attend moving such an asteroid into orbit around the Earth, and then systematically mining it—a mountain of high-grade ore a few hundred miles overhead. The value of platinum-group metals alone in a single such world has been estimated as many trillions of dollars—although the unit price would plummet spectacularly if such materials became widely available. Methods of extracting metals and minerals from appropriate asteroids are being studied, for example by John Lewis, a planetary scientist at the University of Arizona.

Some near-Earth asteroids are rich in organic matter, apparently preserved from the very earliest Solar System. Some have been found, by Steven Ostro of the Jet Propulsion Laboratory, to be double, two bodies in contact. Perhaps a larger world has broken in